Exercises on Linear Codes

A Book-Style Reorganization with Fifty Problems

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Preface

These notes reorganize a homework-style collection of exercises on coding theory into a more standard mathematical format. The emphasis is on linear codes over finite fields, with a gradual progression from elementary constructions to duality, minimum distance, cosets, decoding, cyclic codes, and a first glimpse of algebraic-geometric codes.

The problems are intended to be read in order. Exercises 1–40 stay close to the classical core of an introductory course in coding theory. Exercises 41–50 extend the discussion toward more advanced topics.

CHAPTER 1

Exercises on Linear Codes

1. Notation and conventions

Throughout, $\mathbb{F}_q$ denotes the finite field with q elements. A linear code of length n over $\mathbb{F}_q$ is a linear subspace $\mathcal{C} \leq \mathbb{F}_q^n$. If $\dim_{\mathbb{F}_q}(\mathcal{C}) = k$, then $\mathcal{C}$ is called an $[n,k]_q$*-code*. Its minimum Hamming distance is denoted by $d(\mathcal{C})$, and when this value is specified we write $[n,k,d]_q$.

If $G \in M_{k\times n}(\mathbb{F}_q)$ has row space $\mathcal{C}$, then G is a generator matrix for $\mathcal{C}$. If $H \in M_{(n-k)\times n}(\mathbb{F}_q)$ satisfies

$$\mathcal{C} = \{x \in \mathbb{F}_q^n : Hx^{\mathsf{T}} = 0\},$$

then H is a parity-check matrix for $\mathcal{C}$. The dual code is

$$\mathcal{C}^{\perp} = \{y \in \mathbb{F}_q^n : \langle x, y\rangle = 0 \text{ for all } x \in \mathcal{C}\}.$$

All vector spaces are over the ground field indicated in the statement.

2. Basic constructions and systematic form

Exercise 1 (A single-erasure design problem)**.** Consider a communication channel over $\mathbb{F}_3$ that takes as input a vector of length $n = 5$ and erases exactly one coordinate, without revealing in advance which coordinate will be erased. Construct two distinct linear $[5,4]_3$-codes that allow every message of length 4 to be recovered from the received word.

Exercise 2 (The repetition code)**.** Let $n \geq 1$, and let $\mathcal{C} \subseteq \mathbb{F}_q^n$ be the repetition code

$$\mathcal{C} = \{(a, a, \ldots, a) : a \in \mathbb{F}_q\}.$$

Determine its length, dimension, cardinality, and minimum distance. Write down a generator matrix and a parity-check matrix.

Exercise 3 (The single parity-check code)**.** Let

$$\mathcal{C} = \Big\{(x_1, \ldots, x_n) \in \mathbb{F}_q^n : \sum_{i=1}^{n} x_i = 0\Big\}.$$

Show that $\mathcal{C}$ is a linear $[n, n-1]_q$-code. Determine its minimum distance, and exhibit both a generator matrix and a parity-check matrix.

Exercise 4 (Systematic form over $\mathbb{F}_5$)**.** Let $\mathcal{C}$ be the code generated over $\mathbb{F}_5$ by

$$G = \begin{bmatrix} 1 & 2 & 4 & 0 & 1 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 0 & 0 & 3 & 1 \end{bmatrix}.$$

Use elementary row operations to rewrite G in systematic form.

Exercise 5 (Enumerating codewords over two fields)**.** Let $\mathcal{C}$ be the code generated by

$$G = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

over $\mathbb{F}_q$.

(1) For $q = 2$, determine $|\mathcal{C}|$ and list all codewords.
(2) For $q = 4$, determine $|\mathcal{C}|$ and list all codewords, writing $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$ with $\omega^2 = \omega + 1$.

Exercise 6 (Cardinality of a linear code)**.** Let $\mathcal{C} \leq \mathbb{F}_q^n$ be a linear code of dimension k. Prove that $|\mathcal{C}| = q^k$.

Exercise 7 (Systematic encoding). Suppose $G = [I_k \mid A] \in M_{k\times n}(\mathbb{F}_q)$ is a generator matrix in systematic form. Show that every information word $u \in \mathbb{F}_q^k$ produces a unique codeword $uG \in \mathcal{C}$, and that the first k coordinates of uG are precisely the information symbols.

Exercise 8 (A formal proof for systematic parity-check matrices). Let

$$G = [I_k \mid A] \in M_{k\times n}(\mathbb{F}_q).$$

Prove that

$$H = [-A^{\mathsf{T}} \mid I_{n-k}]$$

is a parity-check matrix for the code generated by G.

Exercise 9 (Length, dimension, and generator rank). Let $G \in M_{k\times n}(\mathbb{F}_q)$ be a matrix of rank r. Show that the row space of G is a linear code of length n, dimension r, and cardinality q^r.

Exercise 10 (Puncturing and shortening). Let $\mathcal{C} \le \mathbb{F}_q^n$ be a linear code, and fix a coordinate position i. Define the punctured code $\mathcal{C}^{(i)}$ by deleting the ith coordinate from every codeword. Show that $\mathcal{C}^{(i)}$ is a linear code of length $n-1$, and prove that

$$\dim \mathcal{C} - 1 \le \dim \mathcal{C}^{(i)} \le \dim \mathcal{C}.$$

3. Duality and self-duality

Exercise 11 (A parity-check matrix in systematic form). Consider the code generated by

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

Compute a parity-check matrix in systematic form.

Exercise 12 (Self-orthogonal or self-dual?). Let $\mathcal{C} \le \mathbb{F}_2^8$ be the code generated by

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 \end{bmatrix}.$$

Determine whether $\mathcal{C}$ is self-orthogonal and whether it is self-dual. Decide whether the answer changes if one replaces $\mathbb{F}_2$ by $\mathbb{F}_p$, where p is an odd prime.

Exercise 13 (A necessary condition for self-duality). Show that if a linear code $\mathcal{C} \le \mathbb{F}_q^n$ is self-dual, then n is even and $\dim \mathcal{C} = n/2$.

Exercise 14 (The tetracode from a parity-check matrix). Let $\mathcal{C}_1$ be the code over $\mathbb{F}_3$ with parity-check matrix

$$H_1 = \begin{bmatrix} 2 & 2 & 1 & 0 \\ 0 & 1 & 1 & 2 \end{bmatrix}.$$

Determine whether $\mathcal{C}_1$ is self-orthogonal and whether it is self-dual.

Exercise 15 (A code over $\mathbb{F}_4$). Let $\omega \in \mathbb{F}_4$ satisfy $\omega^2 = \omega + 1$, and let $\mathcal{C}_2$ be the code generated by

$$G_2 = \begin{bmatrix} 1 & 0 & 0 & 1 & \omega & \omega \\ 0 & 1 & 0 & \omega & 1 & \omega \\ 0 & 0 & 1 & \omega & \omega & 1 \end{bmatrix}.$$

Determine whether $\mathcal{C}_2$ is self-orthogonal and whether it is self-dual.

Exercise 16 (Weight and self-inner product over $\mathbb{F}_3$). For $x \in \mathbb{F}_3^n$, prove that

$$\mathrm{wt}(x) \equiv \langle x, x\rangle \pmod 3.$$

Exercise 17 (Doubly-even type argument)**.** Let $\mathcal{C} \leq \mathbb{F}_2^n$ be a self-orthogonal code having a generator matrix whose rows all have Hamming weight divisible by 4. Prove that every codeword of $\mathcal{C}$ has Hamming weight divisible by 4.

Exercise 18 (Dual of a sum)**.** For linear codes $\mathcal{C}_1, \mathcal{C}_2 \leq \mathbb{F}_q^n$, prove that

$$(\mathcal{C}_1 + \mathcal{C}_2)^\perp = \mathcal{C}_1^\perp \cap \mathcal{C}_2^\perp.$$

Exercise 19 (Dual of an intersection)**.** For linear codes $\mathcal{C}_1, \mathcal{C}_2 \leq \mathbb{F}_q^n$, prove that

$$(\mathcal{C}_1 \cap \mathcal{C}_2)^\perp = \mathcal{C}_1^\perp + \mathcal{C}_2^\perp.$$

Exercise 20 (Dimension formula)**.** Let $\mathcal{C} \leq \mathbb{F}_q^n$ be a linear code. Prove that

$$\dim \mathcal{C} + \dim \mathcal{C}^\perp = n.$$

Deduce that $\mathcal{C} \subseteq \mathcal{C}^\perp$ if and only if $\dim \mathcal{C} \leq n/2$.

4. Minimum distance and parity-check geometry

Exercise 21 (Distance of the single parity-check code)**.** Using only the parity-check matrix $H = [1\ 1\ \cdots\ 1]$, determine the minimum distance of the $[n, n-1]_q$ single parity-check code.

Exercise 22 (Distance of the tetracode)**.** The tetracode over $\mathbb{F}_3$ is

$$\mathcal{C} = \{(x_1, x_2, x_1 + x_2, x_1 - x_2) : x_1, x_2 \in \mathbb{F}_3\}.$$

Using a parity-check matrix for this code, determine its minimum distance.

Exercise 23 (A binary $(7,4)$ code)**.** Let $\mathcal{C} \leq \mathbb{F}_2^7$ be generated by

$$G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Determine the minimum distance of $\mathcal{C}$.

Exercise 24 (A binary $(6,3)$ code)**.** Let $\mathcal{C} \leq \mathbb{F}_2^6$ be generated by

$$G = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}.$$

Determine the minimum distance of $\mathcal{C}$, and decide whether $\mathcal{C}$ contains a codeword of weight 4.

Exercise 25 (Distance from a parity-check matrix)**.** Let $\mathcal{C} \leq \mathbb{F}_q^n$ have parity-check matrix H. Prove that $d(\mathcal{C})$ is the smallest number of columns of H that are linearly dependent.

Exercise 26 (Equivalent definitions of minimum distance)**.** Let $\mathcal{C} \leq \mathbb{F}_q^n$ be a nonzero linear code. Prove that

$$d(\mathcal{C}) = \min\{\mathrm{wt}(c) : 0 \neq c \in \mathcal{C}\}.$$

Exercise 27 (Error-correcting capability)**.** Let $\mathcal{C}$ be a code of minimum distance d. Prove that every Hamming ball of radius

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor$$

centered at a codeword is disjoint from every other such ball.

Exercise 28 (Perfect repetition codes)**.** Let $\mathcal{C} \subseteq \mathbb{F}_2^n$ be the binary repetition code of length n. Determine for which values of n this code is perfect.

Exercise 29 (Puncturing a binary $(24,12,8)$ code)**.** Assume that a binary $[24,12,8]$ code exists. Does puncturing it in one coordinate yield a perfect code? Justify your answer using the sphere-packing bound.

Exercise 30 (Extending a binary $(5,3,3)$ code)**.** Suppose a binary $[5,3,3]$ code is extended by one coordinate. Determine whether the resulting code can be perfect. Give a geometric interpretation of perfect codes in terms of Hamming spheres.

5. Cosets, syndromes, and decoding

Exercise 31 (Disjoint Hamming spheres in the tetracode)**.** Let $\mathcal{C}$ be the tetracode over $\mathbb{F}_3$:

$$\mathcal{C} = \{(x_1, x_2, x_1 + x_2, x_1 - x_2) : x_1, x_2 \in \mathbb{F}_3\}.$$

Construct the Hamming spheres of radius 1 around the codewords of $\mathcal{C}$ and prove that they are pairwise disjoint.

Exercise 32 (A non-linear subset inside a linear code)**.** Let $\mathcal{C} \leq \mathbb{F}_2^6$ be generated by

$$G = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

Decide whether the set of codewords of $\mathcal{C}$ whose weights are divisible by 4 forms a linear code.

Exercise 33 (A coset-weight construction)**.** Find a nonzero binary linear $[4,k]$ code $\mathcal{C}$ such that every coset of $\mathcal{C}$ contains a unique vector of minimum weight, while some coset has weight strictly greater than

$$\left\lfloor \frac{d(\mathcal{C}) - 1}{2} \right\rfloor.$$

Exercise 34 (Syndrome decoding for the tetracode)**.** Let $\mathcal{C}$ be the tetracode over $\mathbb{F}_3$. Decode the received vectors

$$(1,1,1,1), \qquad (1,-1,0,-1)$$

by means of a syndrome table.

Exercise 35 (Cosets of a binary $(7,4)$ code)**.** For the code of Exercise 23:

(1) determine the number of cosets,
(2) construct a complete table of syndromes and coset leaders,
(3) decode the received vectors $(1,1,0,1,0,1,0)$ and $(0,0,1,0,0,0,0)$.

Exercise 36 (A systematic analysis of a binary $(6,3)$ code)**.** For the code of Exercise 24:

(1) rewrite the generator matrix in systematic form,
(2) compute a parity-check matrix,
(3) decide whether the code is self-orthogonal or self-dual,
(4) determine the number of cosets.

Exercise 37 (Erasure decoding)**.** For the binary $[6,3]$ code of Exercise 24, suppose the decoder receives

$$(1,0,0,0,*,1),$$

where $*$ denotes an erasure. Determine the transmitted codeword.

Exercise 38 (A coset of weight 2)**.** For the binary $[6,3]$ code of Exercise 24, identify a coset of weight 2 and determine all of its coset leaders.

Exercise 39 (When is syndrome decoding unambiguous?)**.** Let $\mathcal{C} \leq \mathbb{F}_q^n$ have minimum distance d. Prove that every coset has at most one leader of weight at most $\lfloor (d-1)/2 \rfloor$. Explain why this is the theoretical basis of unique decoding up to the correction radius.

Exercise 40 (Syndromes and cosets)**.** Let H be a parity-check matrix of a linear code $\mathcal{C} \leq \mathbb{F}_q^n$. Prove that two words $x, y \in \mathbb{F}_q^n$ have the same syndrome if and only if they lie in the same coset of $\mathcal{C}$.

6. Further exercises and extensions

Exercise 41 (Monomial equivalence and duals)**.** Give an example of two codes $\mathcal{C}_1$ and $\mathcal{C}_2$ over the same field, together with a monomial matrix M, such that

$$\mathcal{C}_1 M = \mathcal{C}_2 \qquad \text{but} \qquad \mathcal{C}_1^{\perp} M \neq \mathcal{C}_2^{\perp}.$$

Exercise 42 (A binary weight identity)**.** If $x, y \in \mathbb{F}_2^n$, let $x \cap y$ denote the vector whose support is the intersection of the supports of x and y. Prove that

$$\operatorname{wt}(x+y) = \operatorname{wt}(x) + \operatorname{wt}(y) - 2\operatorname{wt}(x \cap y).$$

Exercise 43 (The ternary Golay code G_{12})**.** Let G_{12} be the ternary code generated by $[I_6 \mid A]$, where

$$A = \begin{bmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{bmatrix}.$$

Show that G_{12} is a self-dual $[12, 6, 6]_3$-code. Then explain why puncturing an appropriate coordinate produces an $[11, 6, 5]_3$ code.

Exercise 44 (Puncturing, extending, and even-like codewords)**.** Let $\mathcal{C} \leq \mathbb{F}_q^n$ be a linear code, and let $\mathcal{C}_1$ be obtained by puncturing the rightmost coordinate and then extending the punctured code by appending the negative sum of the remaining coordinates.

(1) Prove that $\mathcal{C} = \mathcal{C}_1$ if and only if every codeword of $\mathcal{C}$ has coordinate sum 0.
(2) Prove that if $\mathcal{C}$ is self-orthogonal and contains the all-one vector, then $\mathcal{C} = \mathcal{C}_1$.

Exercise 45 (A first Reed–Solomon exercise)**.** Let $\alpha_1, \ldots, \alpha_n \in \mathbb{F}_q$ be distinct, and let $1 \leq k \leq n$. Define

$$\mathrm{RS}_k(\alpha_1, \ldots, \alpha_n) = \{(f(\alpha_1), \ldots, f(\alpha_n)) : f \in \mathbb{F}_q[x],\ \deg f < k\}.$$

Show that this is a linear $[n, k]_q$-code.

Exercise 46 (Reed–Solomon codes are MDS)**.** With the notation of Exercise 45, prove that

$$d(\mathrm{RS}_k(\alpha_1, \ldots, \alpha_n)) = n - k + 1.$$

Exercise 47 (Cyclic codes as ideals)**.** Assume $\gcd(n, q) = 1$. Let $R = \mathbb{F}_q[x]/\langle x^n - 1\rangle$, and let σ denote the cyclic shift on $\mathbb{F}_q^n$:

$$\sigma(c_0, c_1, \ldots, c_{n-1}) = (c_{n-1}, c_0, \ldots, c_{n-2}).$$

Show that the linear codes $\mathcal{C} \leq \mathbb{F}_q^n$ that are invariant under σ correspond exactly to ideals of the quotient ring R.

Exercise 48 (Invariant subspaces and the quotient-ring model)**.** Make explicit the correspondence of Exercise 47 by sending

$$(c_0, c_1, \ldots, c_{n-1}) \longmapsto c_0 + c_1 x + \cdots + c_{n-1} x^{n-1} \in R.$$

Prove that invariance under cyclic permutation is equivalent to closure under multiplication by x modulo $x^n - 1$.

Exercise 49 (A binary Goppa code identity)**.** Let $L = \{\alpha_1, \ldots, \alpha_n\} \subseteq \mathbb{F}_{2^m}$, and let $g(x) \in \mathbb{F}_{2^m}[x]$ be squarefree with $g(\alpha_i) \neq 0$ for all i. The classical binary Goppa code $\Gamma(L, g)$ may be described as

$$\Gamma(L, g) = \Big\{ c = (c_i) \in \mathbb{F}_2^n : \sum_{i=1}^{n} \frac{c_i}{x - \alpha_i} \equiv 0 \pmod{g(x)} \Big\}.$$

Show that

$$\Gamma(L, g) = \Gamma(L, g^2).$$

Exercise 50 (A first algebraic-geometric coding exercise)**.** Let X be a smooth projective curve of genus g over $\mathbb{F}_q$, let

$$D = P_1 + \cdots + P_n$$

be a sum of distinct rational points on X, and let G be a divisor whose support is disjoint from that of D. Define the evaluation code

$$C_L(D, G) = \{(f(P_1), \ldots, f(P_n)) : f \in L(G)\}.$$

(1) Using the Riemann–Roch theorem, prove that if $\deg G < n$, then

$$\dim C_L(D, G) \geq \deg G + 1 - g.$$

If in addition $2g - 2 < \deg G < n$, prove that equality holds.

(2) Prove that every nonzero codeword of $C_L(D, G)$ has at most $\deg G$ zero coordinates. Deduce that

$$d\big(C_L(D, G)\big) \geq n - \deg G.$$

Concluding remark

This set is intended as a bridge between computational fluency and conceptual structure. The first forty problems may be approached directly with row reduction, parity-check matrices, and elementary counting arguments. The final ten problems indicate how the same language continues into Reed–Solomon, cyclic, Goppa, and algebraic-geometric codes.